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Evaluating the use of grey-box system identification for digital twins in manufacturing automation

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ABSTRACT

A key element of any digital twin is the digital replica, or model, of its physical counterpart. When applied to industrial automation systems, it is important to consider the trade-off between model fidelity and computational complexity when developing this model. In this paper, we investigate the use of grey-box system identification as a means for producing digital twin models, as the approach reduces computational complexity while maintaining model fidelity. Using a three-link robotic manipulator, we evaluate the ability the digital twin's system identification module to produce a high-fidelity model while under the influence of input disturbances. We then expand these results by evaluating the ability of the determined model to accurately emulate the robotic manipulator for tasks given to digital twins through the system simulation and monitoring modules. The results of this testing demonstrate that the grey-box system identification module is prone to error and sensitive to input disturbances; however, it produces models that are still able to accurately predict the dynamic response of the robotic manipulator. Furthermore, the tests demonstrate the determined models can be used by the digital twin for simulation and monitoring applications with certain limitations.

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KEYWORDS
Digital twin; system identification; grey-box modelling

1. Introduction

Among the many challenges of creating a digital twin (DT) is that of developing the digital replica, or model, of the physical entity of interest. This model must achieve a minimum level of detail (fidelity) to approximate the physical system. As such, the model developer's task is to find an optimum balance between model accuracy and computation cost.

In the case of intelligent manufacturing control, the most useful knowledge often relates to how the system responds to the various inputs. As a result, it is important to be able to model the dynamic response of the system over its range of operating conditions. This field is generally known as system identification and entails the construction of models using sensor data.

Two common approaches used for manufacturing control applications are physics-based modelling and data-driven modelling (Thelen et al. 2022). The former approach can produce a high-fidelity model but requires a thorough understanding of the underlying physics. As well, this approach often results in a model that is too complex and too computationally expensive for practical DT applications. Data-driven modelling provides an alternative that is less computationally complex but is typically of lower fidelity than a physics-based model.

In this paper, we investigate a DT modelling approach that combines physics-based and data-driven models for improved fidelity, otherwise known as grey-box modelling. We begin with some background on DTs for manufacturing automation, then describe the robotic manipulator used in this work as well as short overview of the DT and its modules. We then begin evaluating the grey-box system identification approach by first testing its ability to accurately determine physical system parameters, and then its ability to effectively emulate the dynamic system response. Then we proceed to use models

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determined by the proposed approach in real DT applications to further examine the usefulness of the approach. Finally, we present our conclusions and future work building on the results of this paper.

2. Background

In the area of manufacturing automation, the digital twin has been demonstrated by multiple researchers as an effective means of realising flexible intelligent production systems (Kritzinger et al. 2018; Abisset-Chavanne et al. 2024). For example, He et al. (2019) applied a DT to detect failures during the control of an industrial process, and demonstrated they could use the DT to recover from the failure, re-optimise for the new operating conditions, and continue production. Liu et al. (2020) applied a DT of a workshop involving human and machine elements to optimise production output through scheduling, running multiple simulations to find an optimal schedule before implementing it in the workshop. Vachalek et al. (2017) developed a DT of a small pneumatic cylinder production line, optimising the entire production line and reducing production time by 5.2%. Zhao et al. (2019) applied a DT to a micro-punching system, increasing the accuracy from ± 2 micrometre to ± 1 micrometre while also more than doubling the rate of punching. In addition to real applications researchers like Jeon and Schuesslbauer (2020) have proposed and argued for the use of DTs in flexible production systems, stating digital twins may assist in the optimisation of workshop flow and management. Our work in this area has focused on the application of DTs to distributed intelligent sensing and control systems (DISCS) (Lesage and Brennan 2022).

By definition, a digital twin is a 'virtual model designed to accurately reflect a physical object' (IBM 2022). The DT combines sensor data with a digital model to monitor, simulate, and analyse the physical system in order to improve its operation (Schleich et al. 2017; Tao et al. 2022). In this section, we begin with an overview of the recent work on the creation and implementation of digital twins for industrial applications. We then focus more closely on the modelling approaches used in digital twin design for system identification.

2.1. Creation and implementation of digital twins

The various applications of digital twins in industry demonstrate their immense potential to revolutionise smart manufacturing. However, it still remains unclear how to create or implement one, and no standard yet exists. The International Organization for Standardization (ISO) is attempting to address this with their upcoming standard 23247 (ISO 2021), but, due to the unique nature of individual digital twins, there will likely still be much ambiguity in implementation.

As such, this is an active topic of research, with many attempting to provide the needed framework for digital twins. Wu et al. (2021) examined this challenge at that conceptual stage. They put forward a five step framework intended to capture the major functional aspects of the digital twin, and guide engineers through the process. They went on to apply this in a case study of a smart electric vehicle.

Leng et al. (2019) attempted to tackle the challenge of creating a digital twin of a smart workshop. They proposed a detailed framework for this digital twin, which includes a network of digital twins of various CPPS's within the workshop. Their goal behind creating this is to allow engineers to monitor, control, and eventually optimise the performance of a smart workshop. They went on to deploy a prototype using this framework within Java.

Zhuang et al. (2021) recently published a paper proposing a five-step framework for making a digital twin workshop, with the aim of predicting equipment failures. To do this, the incorporated Markov Chains into their digital twin, and gathered statistical data surrounding the performance of workshop equipment. This feature permitted them to predict failure and act early on, thus allowing for preventative maintenance and minimizing down time of machinery.

Qamsane et al. (2021) created a practical method for engineers to follow in the design of digital twins, which they used for their bearing failure detection. Their process involves determining if a digital twin is an effective solution, evaluating the resources and requirements for the twin, designing, deploying, and testing it. Their approach provides a route for engineers to evaluate the applicability of digital twin solutions.

In addition to the challenge of creating digital twins for manufacturing, there exist a multitude of

research into the technicalities of implementing them. T. H. Uhlemann, C. Lehmann, and R. Steinhilper (2017) attempted to address a practical method for implementing a digital twin in small and medium sized enterprises. They proposed the use of machine vision and sensor based tracking as an economical, practical, and effective method for gathering the data necessary to generate the digital twin of a workspace. They aim to make it possible for smaller organizations to apply Industry 4.0 concepts, and thus spread the use of digital twins.

Biesinger et al. (2019) examined the challenge of automatically generating a digital twin of a CPPS. They determined one of the key challenges to be data transfer to and from the physical twin, which they proposed solving using a technology called the manufacturing service bus. This provides high data transfer speeds with minimal delay. By applying this technology, they succeeded in rapidly gathering data from an industrial programmable logic controller (PLC), providing the basis to begin automatic generation.

Talkhestani et al. (2020) proposed a methodology to automatically synchronise a CPPS with a digital twin. They refer to this as the anchor point method, which permits synchronization based on PLC code analysis. With this synchronised digital twin, they rapidly tested new configurations of the CPPS without having to undertake the traditional methods of reconfiguration. In their case study, they estimated this approach could reduce the time required for the reconfiguration process by 58%.

Borangiu et al. (2020) proposed an architecture for a smart manufacturing system using three layers of digital twins. At the lowest layer, the digital twins are responsible for collecting and transmitting data to the physical system. The second layer of digital twins processes the data on performance, maintenance, and detecting anomalies. In the final layer, intelligence is achieved, and these digital twins possess machine learning and AI algorithms to optimise and reconfigure the CPPS as required. This layered approach permits the system to effectively collect and process data, as well as make decisions on it, without exceeding computing limitations. The researchers went on to implement this in a robotic grabber system.

Xia, Lu, and Zhang (2020) attempted to provide a more complete structure for the digital twin workshop initially proposed by Tao et al. (2019). Their implementation included a digital physical workshop, a digital twin engine, and the virtual workshop. The digital physical workshop is the actual workshop where data is collected and devices and personnel are interacted with, while the digital twin engine is responsible for processing and analysing data as well as interacting with the workshop. The virtual workshop uses this data to simulate, optimise, and visualise the physical workshop. Similar to the work of Borangiu et al. (2020), this implementation represents a layered approach.

Preuveneers, Joosen, and Ilie-Zudor (2018) addressed a critical issue regarding CPPS which contain multiple digital twins, of which many do. They note that as functionality between various digital twins overlaps, there is a potential for data to conflict and result in system failures. To address this, they proposed using feature toggles and software circuit breakers, thus containing a failure should it occur and permit the rest of the system to continue operating. The end result is a more robust CPPS.

Yu-Ming, Bing, and San-Peng (2020) addressed another key issue regarding data from digital twins during implementation. When a single digital twin collects data, it typically does so multiple sources, and as such the data is heterogeneous. Recognizing this, the authors applied AI techniques to sort, process, and interpret the heterogeneous data. With this capability, they created a digital twin of a robotic sorting arm which could simulate and optimise the physical twin. In line with the definitions presented by other authors, their digital twin communicated with the manufacturing execution system to implement optimal performance.

Ait-Alla et al. (2019) went to one of the more fundamental questions surrounding sensor requirements for a digital twin. They used a sample CPPS, which consisted of a series of moving carts transferring components between manufacturing stations, for their experiment. By increasing the number of sensors in their system, they would slow communication between the digital and physical twins. However, they more rapidly detected malfunctions in the system as a result, and could more quickly redirect

resources when required. The net effect was an overall increase in productivity in configurations with more sensors.

The concept of visualizing a digital twin has been another topic of investigation, with Zhu, Liu, and Xu (2019). demonstrating the use of augmented reality for this. They used Microsoft's HoloLens to visualise the data from the digital twin of a CNC milling machine in real time. The use of this environment permits users to view the operational status of the CNC mill and control it as required, thus creating an HMI. They intend to use this application to further develop intelligent control processes through augmented reality in the future. In the domain of geographical scene modelling, J. Zhu et al. (2024) propose a data- and knowledge-driven approach to enhance the ability of virtual geographical scenes to describe the changing real world. This approach facilitated the effective use and re-use of data.

The literature makes two concepts abundantly clear. The first is that digital twins are a key element of smart manufacturing systems. Researchers have already established the ability of digital twin's to detect faults (Azangoo, Taherkordi, and Blech 2020), provide maintenance predictions (Qamsane et al. 2021), dynamically reconfigure control systems (Zhang et al. 2020), optimise production systems (Liu et al. 2020; Zhao et al. 2019; Vachalek et al. 2017), and assist workers in factories (Meier et al. 2021). All of these are critical elements in maximizing the efficiency of production and manufacturing, as well as automating various processes.

The second is that there remains a great amount of work in determining how to design and implement digital twins for smart manufacturing systems. It is clear that no one standard method exists for creating digital twins, and as such various authors proposed different methods with which this technology should be designed and implemented. In this paper, we focus on the model of the physical system (i.e. 'system identification') given the key role that it plays in the creation and implementation of digital twins.

2.2. System identification for digital twins

In flexible automation systems, the physical model should be capable of capturing the dynamic behaviour of the system while accommodating noise (e.g. sensor noise, unmodelled parameters).

In general, two approaches are used to obtain a model of the physical system. The first, physics-based modelling, requires full knowledge of the system to construct a rigorous 'white box' model. However, as is often the case, full knowledge of the physical system cannot be obtained: as a result, this approach often results in a reduced order model with correspondingly reduced capabilities. The second approach, data-driven modelling, uses statistical identification techniques to construct a 'black-box' model of the system. However, black-box models typically require a substantial amount of data to achieve model accuracy; as well, these models are typically in a form where the model parameters do not necessarily have a direct relation to the physical system, making the model literally a 'black-box' to the user (Reynders, Diriken, and Saelens 2014).

Table 1 provides a summary of the literature in this context: i.e. whether the work of these papers considered the two key modelling feature of digital twins, those being a physics based model and/or a data based model in their implementation.

To address these issues, a third approach, 'grey-box' modelling, has been proposed that combines the advantages of these two approaches (Tulleken 1993; Bohlin and Graebe 1994). Grey-box models employ the known, or believed, physics of the physical system along with collected sensor data. A model is then constructed by estimating the various parameters of the physics-based model. The combination of estimation and physical modelling that grey-box modelling offers makes it a natural choice for DTs in a stochastic manufacturing environment.

A grey-box modelling approach is well-suited to this application area given its focus on addressing three common difficulties when modelling industrial processes (1): stochastic driving noise (2), nonlinear state equations, and (3) unknown modelling parameters (Bohlin and Graebe 1994). To meet these challenges, the grey-box modelling approach considers partially known models. More specifically, the model is structured in the standard format of inputs, outputs, and system parameters; however, the model's system parameters include both a set of known constants, c, and a set of unknown constants, α, that are estimated by the grey-box model. Compared to black-box models, grey-box models have been shown to yield fewer and more physically meaningful parameters that are valid over a wide-range of operation (Kristensen,

Madsen, and Jørgensen 2004). As well, they have proven to be more robust to noise than white-box models (Bohlin and Graebe 1994).

Recent work in this area span a wide range of DT applications domains. For example, E. Atasayan, E. Milanov, and A. Dursun Alkan (2024) propose a grey-box model to investigate the feasibility of determining ship manoeuvring dynamics using system identification methods, and to evaluate the manoeuvring characteristics using experimental results compared to predicted results. Liu et al. (2023) propose a grey-box model of optical-network capacity to support more efficient network control and management. Morkunaite et al. (2024) propose a grey-box modelling approach for multistory residential building thermal dynamics. They note that commonly used physics-based white box models are time-consuming and require vast expertise. On the contrary, black box forecasting models, which rely primarily on building energy consumption data, lack fundamental insights and hinder re-use.

In the next sections we propose a digital twin design approach that utilises grey-box modelling for system identification. We begin with a description of the test case in Section 3 then provide an overview of the digital twin design and testing in Sections 4–7.

Table 1. Digital twin modelling approaches.

Author(s)	Physics-basedModel[a]	Data-drivenModel[a]
Ait-Alla et al. (2019)	?	?
Biesinger et al. (2019)	?	Yes
Borangiu et al. (2020)	?	Yes
He et al. (2019)	Yes	Yes
Jeon and Schuesslbauer (2020)	?	?
Leng et al. (2019)	Yes	Yes
Liu et al. (2020)	Yes	Yes
Meier et al. (2021)	Yes	Yes
D. Preuveneers, W. Joosen, and E. Ilie-Zudor (2018)	?	Yes
Qamsane et al. (2021)	Yes	Yes
Talkhestani et al. (2020)	?	Yes
Tao et al. (2019)	No	Yes
T. H. Uhlemann, C. Lehmann, and R. Steinhilper (2017)	No	Yes
Vachalek et al. (2017)	Yes	Yes
Wu et al. (2021)	No	Yes
Xia, Lu, and Zhang (2020)	Yes	Yes
Q. Yu-Ming, X. Bing, and D. San-Peng (2020)	Yes	Yes
Zhang et al. (2020)	Yes	Yes
Zhao et al. (2019)	Yes	Yes
Z. Zhu, C. Liu, and X Xu (2019).	Yes	Yes
Zhu et al. (2024)	No	Yes
Zhuang et al. (2021)	Yes	Yes

[a]Note that a '?' denotes if it was uncertain from the paper if the authors considered that feature.

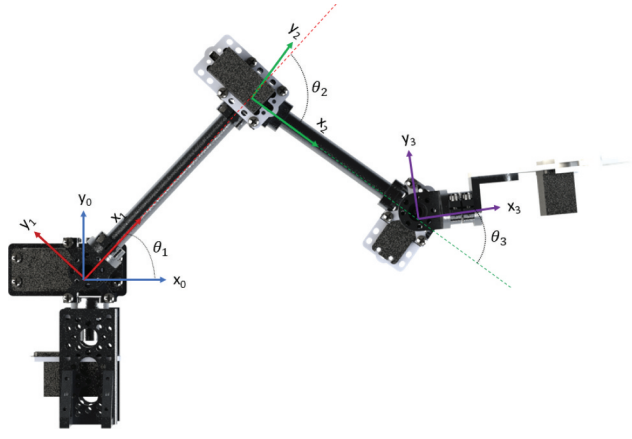


Figure 1. The robotic manipulator used in this work. Note that the various local coordinate systems of each link are defined relative to that of the previous link.

3. Overview of the robotic manipulator

The three-link planer robotic manipulator shown in Figure 1 was used to evaluate our digital twin modelling approach. The choice to use a simplified manipulator configuration (rather than a full-scale industrial robot) was taken for two main reasons. First, an accurate physics-based model of the experimental system was required to compare with our proposed grey-box model. This lower-order model allowed a physics-based model to be developed that could be validated over a range of operating conditions (details of the validation process and results can be found in (Lesage and Brennan 2022)). Secondly, the manipulator shown in Figure 1 effectively consists of three control loops. As each link possesses a servo motor which controls and measures its position relative to the other links, this manipulator provided a suitable platform to allow us to focus on application of digital twins to distributed control problems.

To simulate a real time distributed control system, each joint is viewed as a separate control loop that must coordinate to achieve the overall path planning. The governing physics of the manipulator are presented in Eq. 1. Here, τ is a vector containing the torques applied at the joints by the servo motors, $M(\theta)$ is a mass matrix, $B(\theta,\theta)$ is a matrix containing the Coriolis and centrifugal forces, $G(\theta)$ is vector containing the gravitational forces acting on the links, and D is a diagonal matrix containing viscous damping constants.

$$\tau = M(\theta)\,\ddot{\theta} + B\left(\theta,\dot{\theta}\right) + G(\theta) + D\dot{\theta} \tag{1}$$

Two simplifying assumptions are used for the manipulator joints. First, each arm possesses an absolute encoder with infinite resolution. This eliminates accuracy issues surrounding resolution of the encoder, and eliminates issues with calibrating the sensors about a new location. Second, it is assumed that the resolution provided by the digital to analog conversion, and vice versa, is so fine that the effects of converting the digital value to an analogue voltage in the motor may be neglected. Lastly, outside inherent sensor accuracy and digital analog conversion are not the only issues that occur in measurement, others include noise, sensor offset, etc. However, no assumptions are made regarding noise or offset, as these are left as independent variables for testing.

Each servo motor does not receive a torque command but rather a voltage signal from the controller. Eq. 2 presents the linear equation defining this relationship where V_c is a voltage constant, and Vin is the input voltage. This relationship simplifies the dynamics of the servo while also accounting for the key dynamics of interest. Of final note on the servo motors, a saturation limit of ±8 V exists.

$$\tau = V_c V_{in} \tag{2}$$

These governing equations require a total of eighteen physical constants (provided in Appendix). These constants include moments of inertia, centres of gravity, masses, voltage constants, link lengths, and damping constants.

In addition to the governing dynamics of the manipulator, the control strategy must be defined. This will assist in evaluating the efficacy of the determined model later on. The manipulator is controlled via a computed torque controller which follows a cubic trajectory for each link defined by Eq. 3. Here $\theta_d(t)$ is the desired link angle at a given time instant, a_i are constants, and t is the time along the path beginning when the motion commences. The target time is the desired duration of the motion.

$$\theta_d(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 \tag{3}$$

4. Overview of the digital twin

In this section we provide a brief overview of the DT used in this work to demonstrates how the grey-box modelling approach is implemented as well as how the determined model is applied. We provide a more detailed explanation of the digital twin architecture in our earlier work (Lesage and Brennan 2022), and in this paper we will examine the three modules that constitute the DT software.

The System Identification Module is the first of the three, and is responsible for constructing the grey-box model. Upon a request from the user the module sends a signal through the manipulator, and collects the position data from the encoders in the servo motors. With the collected data it establishes the initial state of three positions and three velocities, the latter with numerical differentiation, and then

undergoes a recursive algorithm to achieve the greatest possible curve fit. The module's graphical user interface is presented in Figure 2.

The System Simulation Module is responsible for applying the determined grey-box model for controller tuning tasks. The module allows the user to pick between pre-set destination and origin locations and simulate the motion of the manipulator under different controller gains and target times. The module's GUI is presented in Figure 3.

The last of the modules, the System Monitoring Module, applies the grey-box model for the task of determining failure of the manipulator. It does this by comparing the measured position of each link to the simulated position of each link at a given instant. If enough consecutive deviations are detected, the module alerts the user that a failure has occurred. The module's GUI is presented in Figure 4.

Each of these modules are innately related to the grey-box modelling approach in some form. As such, examining the performance of all three modules will present insight into the efficacy of the approach for use in DT software.

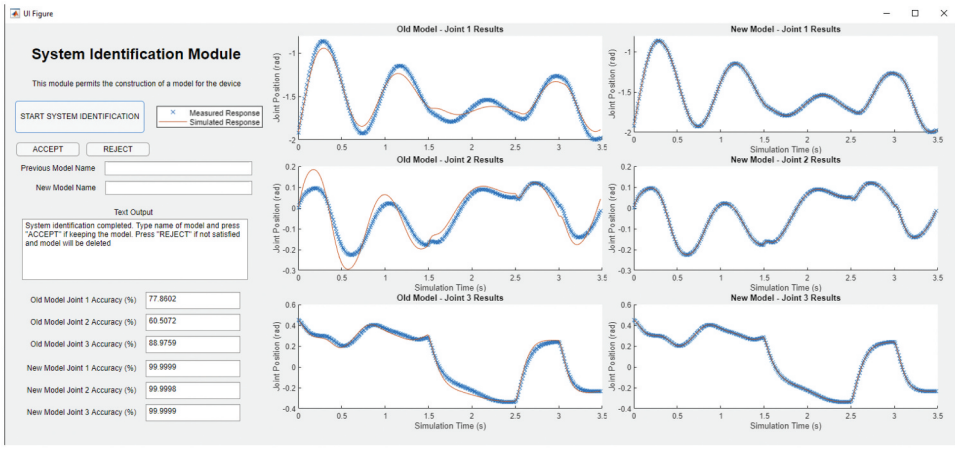


Figure 2. The GUI for the system identification module.

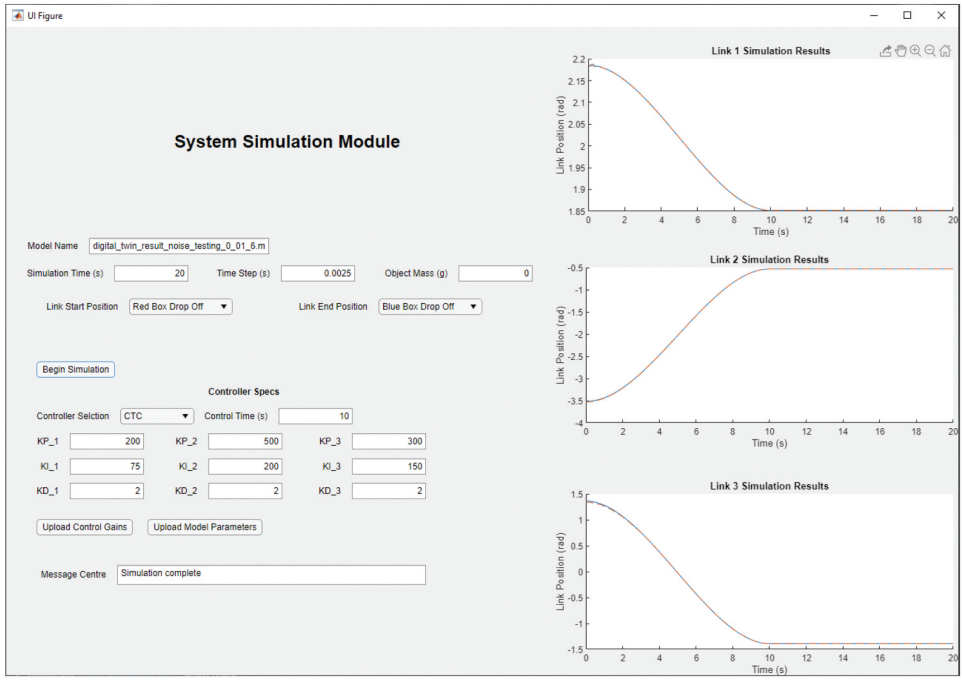


Figure 3. The GUI for the system simulation module.

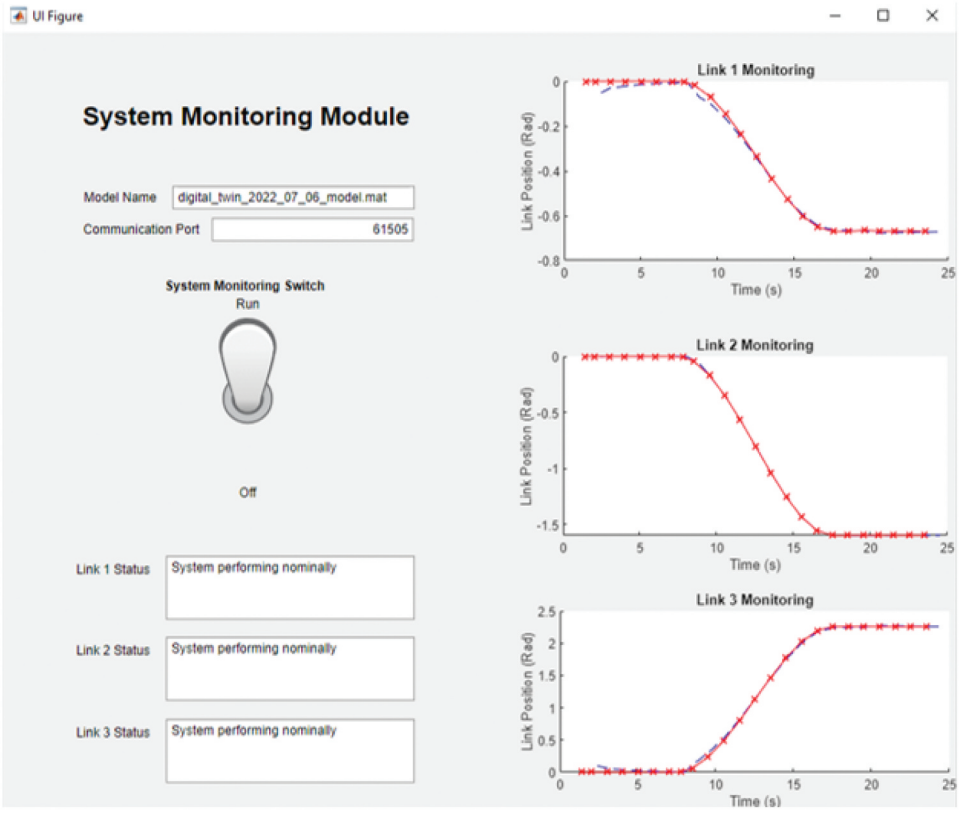


Figure 4. The GUI for the system monitoring module.

5. System identification module

5.1. Testing methodology

The objective in testing the system identification module is to determine the high-fidelity model employed for the other modules. It estimates all the physical parameters contained within Eq. 1. In order to evaluate the grey-box modelling approach through this module, the accuracy of the determined parameters must be established. This can be achieved by calculating a mean error, e_{ij} for each unknown parameter i at link j. Eq. 4 displays this, where α_{ij} is the true value of the parameter and $\widehat{\alpha_{ij}}$ is the estimated parameter.

$$e_{ij} = \frac{\left|\alpha_{ij} - \widehat{\alpha_{ij}}\right|}{\alpha_{ij}} \times 100\% \tag{4}$$

In addition to this evaluation metric, the variance over the tests can be calculated to provide an indication of the precision of the approach. While testing this module, real phenomenon must be taken into account to determine the efficacy of the grey-box modelling approach. In this work, we will examine the effects of sensor noise disturbances, input disturbance through mass uncertainty in the gripper of the manipulator, and the effect of model selection by the user. In all cases, at least ten tests are run.

Sensor noise is established by using a random number generator and a set maximum magnitude of sensor noise. Eq. 5 displays this where θ is measured position, θ is the actual position from the servo encoder, RAND is a random number between 0 and 1, and *MAG* is the noise magnitude. Here, noise magnitudes of 0.001 rad, 0.005 rad, 0.010 rad, and 0.025 rad are used. This is a more strenuous sensor noise model as it also creates an offset compared to noise models that are centred about the true measurement.

$$\hat{\theta} = \theta + RAND \times MAG \tag{5}$$

To model the effects of input disturbances, mass uncertainty is given to an object in the manipulator's gripper. The terms in the equations governing the dynamics allow for such a mass to be present, and by extension the System Identification Module. Usually the module assumes this mass to be zero, but for this set of tests it is assumed that the mass

of the object is 200 g. Three sets of tests will be conducted based on this setting, one where the true mass of the object is 190 g, a second set at 200 g, and a final set at 210 g. These small variations will account for a reasonable amount of mass uncertainty when completing the system identification process with an object in the gripper.

Lastly, the effects of model selection must be accounted for. While the dynamics of the manipulator are entirely deterministic and perfectly known, as the manipulator is a software object, this cannot always be counted on. In the case of a robotic manipulator like this one of the key elements of model uncertainty is the friction model. As such, to further test the limits of the proposed approach a new governing equation is given to the module, that of Eq. 6. Here, μ are Coulombic friction constants and SGN() is the sign function, and as such this new model adds in Coulombic friction terms. The true governing equation, that of Eq. 1, still applies to the manipulator though.

$$\tau = M(\theta)\ddot{\theta} + B(\theta, \dot{\theta}) + G(\theta) + D\dot{\theta} + \mu SGN(\dot{\theta}) \quad (6)$$

5.2. Results and discussion

5.2.1. Sensor noise results

The resulting errors from sensor noise testing are presented in Figure 5, while variance results are presented in Figure 6. For noise magnitudes of 0.001 rad, 0.005 rad, and 0.010 rad twenty simulations were run. However, only eleven simulations were conducted at a noise magnitude of 0.025 rad due to the system identification algorithm's inability to complete with that amount of noise.

The error results show an expected trend of increasing error with increasing noise. However, there are deviations in this trend, and this may be explained by the variance seen in the results. Of all the parameters, the moments of inertia of the links see the largest errors, greater than the other parameters by a significant factor.

5.2.2. Mass error results

The resulting errors from the mass error testing are presented in Figure 7, while variance results are presented in Figure 8. Under all test cases, ten simulations were run.

A similar trend to the noise results is observed in that the moments of inertias error and variances exceed those of the other parameters by a wide margin. Additionally, there is still a non-zero error on multiple parameters even when the true box mass is known exactly.

5.2.3. Model selection results

Unlike the previous tests, the results for model selection are presented in the form of a bar chart. Additionally, the rather than presenting a variance, error bars which encompass the total range in which parameters were determined are given. The results of this testing are given, with error bars in Figure 9(a), Again, ten simulations were ran to produce this result.

While the average errors encountered here are not as large as those encountered during the noise testing, there are still appreciably large errors in the various moments of inertia. Also of interest are the Coulombic friction coefficients determined by the module, shown in Figure 9(b). While the true dynamics have no such friction element present, the module itself determines them to be present in some capacity nonetheless.

5.2.4. Discussion of results

Based on our experiments with the System Identification Module, three key observations can be made for this application. First, there is always some error in the grey-box model's estimation of the unknown parameters, $\widehat{a_{ij}}$. Second, the magnitude of the error in the moments of inertia is significantly larger than the other parameters across all tests. Finally, the variance observed in noise testing is significantly larger compared to the other tests.

Given that DT's are intended to provide a 1:1 representation of their physical counterpart, parameter estimation errors can be problematic. This is potentially even more troubling when examining the results of shown in Figure 7, in which testing that had a 200 g mass in the gripper was effectively a simulation in which no disturbance or modelling error existed. Additionally, the results from the testing with the Coulombic friction model shows that the approach may attribute values to non-existent parameters. As such, it can be stated that the approach is incapable

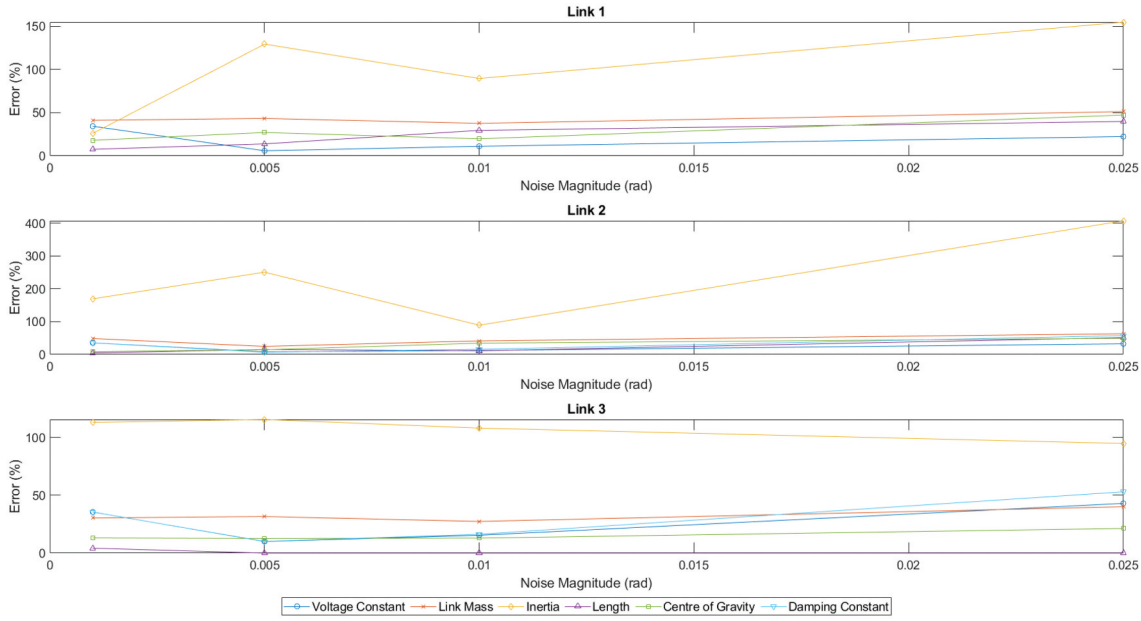


Figure 5. The resulting mean errors over the different noise tests for each noise magnitude.

of perfectly replicating the system, and may even produce an incorrect representation.

Examining the results of the moment of inertia, it is obvious that these parameters have the greatest error in their estimation across all links, across all tests. This may be attributed to their magnitude. All moments of inertia are on the order of $10 - 4$ kg·m^2, which contrast starkly with the other parameters. For example, the mass of links are on the order of 10^{-1} kg. Given the large error, and the large variance seen in this parameter, it suggests that the moment of inertia does not have as strong of an influence on the overall dynamics. As such, the algorithm produces a wider estimate for it.

Finally, the variance in the noise testing must be addressed, as it is much larger compared to other tests. Of additional note is that this variance increases significantly in as the noise magnitude

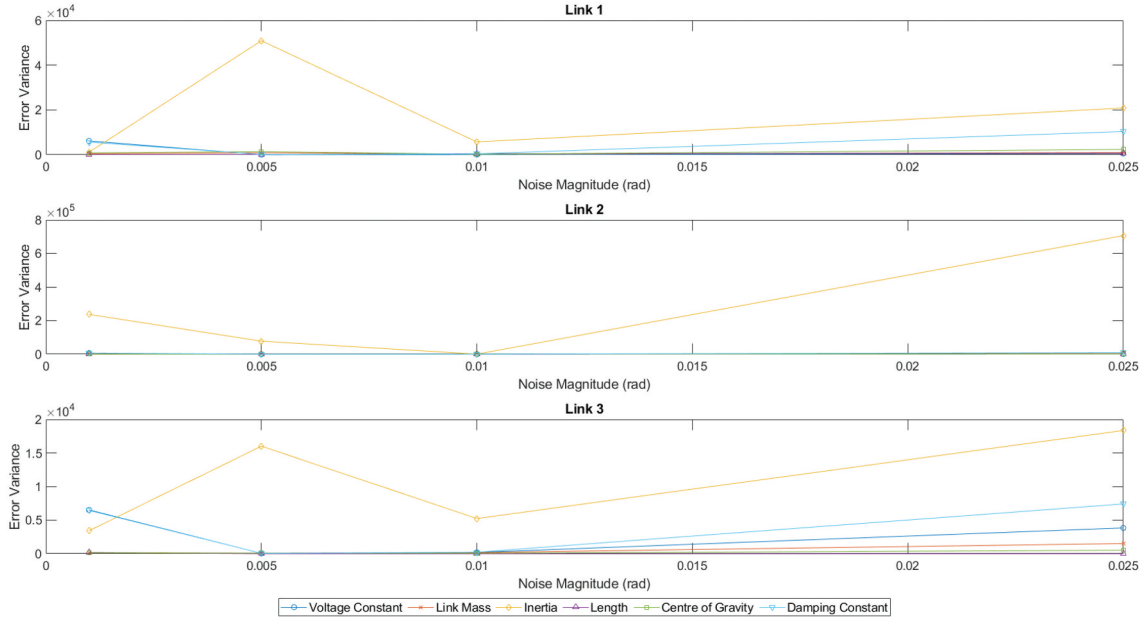


Figure 6. The resulting variances over the different noise tests for each noise magnitude.

increases. This variance may be attributed to the initial estimation of the link velocities. As noted earlier, each servo has a single rotary encoder in it, and as such the module must estimate the initial velocity using numerical differentiation. Despite using a high order differentiation scheme, there are still limitations on how frequently data may be collected. As a result of these limitations, the initial velocity can be greatly misestimated, even at lower levels of noise. Figure 10 demonstrates this phenomenon, where even with the lowest magnitude of noise led to this. The orange line representing the curve fit greatly differs from the measured data at the beginning as a result of an inaccurate velocity calculation, and the algorithm has little hope correcting the curve fit.

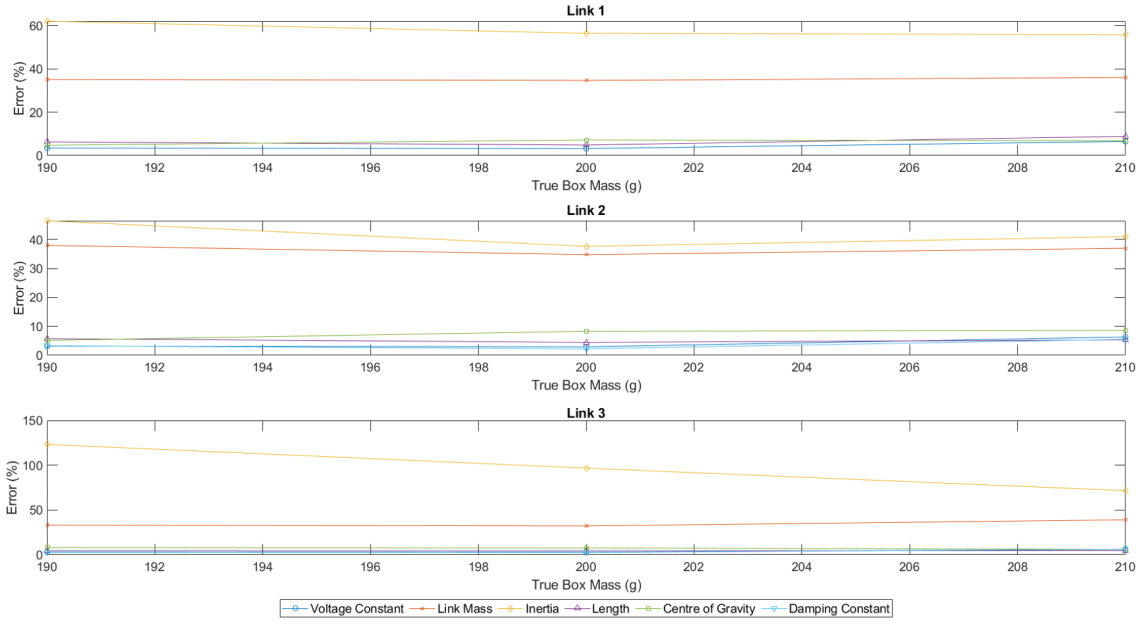


Figure 7. The resulting mean errors over the different true box masses.

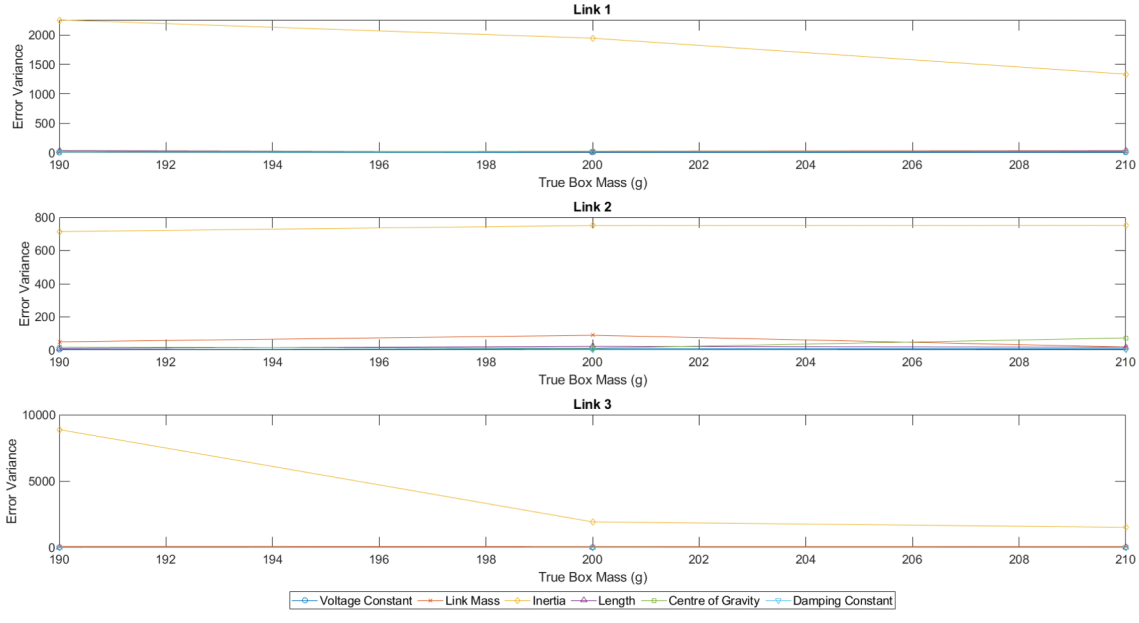


Figure 8. The resulting variances over the different true box masses.

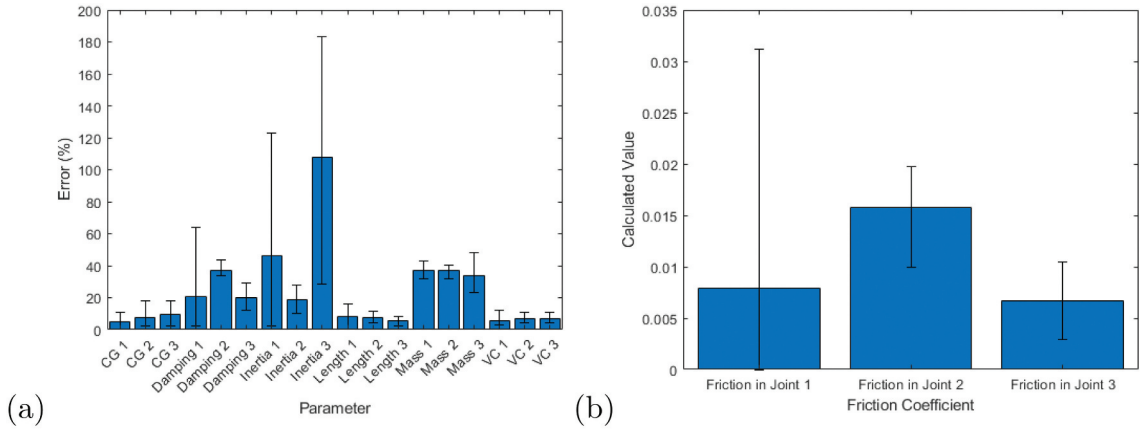


Figure 9. The Coulombic friction model results (a): for the model parameters, *a*, and (b) for the joint Coulombic friction coefficient.

We must now shift the discussion to the ability of the approach to produce a model capable of accurately representing the system. It is evident from these results that the proposed approach is incapable of representing the system perfectly. Some of these sources, like the initial estimate of the velocity, may be mitigated or simply filtered by the developer choosing the DT model. However, sources like model uncertainty can never be fully eliminated. Even if those sources were eliminated, the approach would still not produce a perfect model.

Rather than examining the results in this manner, they can be viewed in the context of Grieves' Test of Virtuality (GTV) (Grieves and Vickers 2017). This is to say the results can be viewed in the context of sensory visual, performance, and reflectivity to determine if the virtual system has the capacity to mirror its physical counterpart. In line with this, we can examine the results of a test in which the Coulombic friction dynamics were present.

Figure 11 displays the results of a test in which the Coulombic friction dynamics were present. The figure

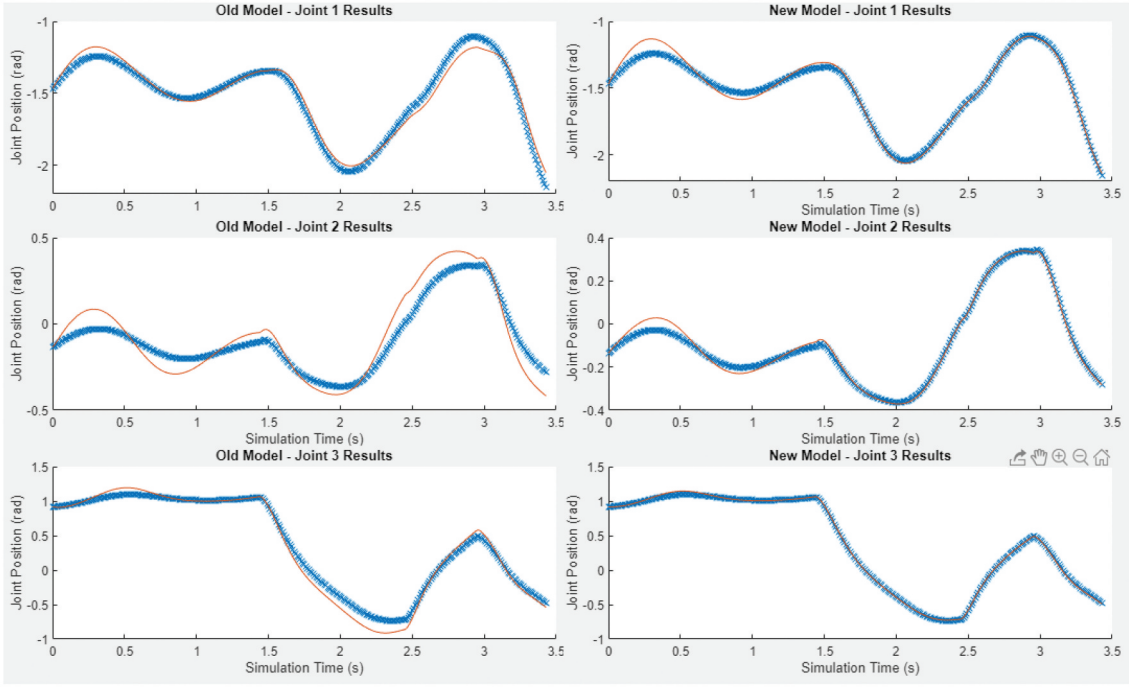


Figure 10. A resulting curve fit when the initial velocity was misestimated. The old model represents the original estimation, and the new model is the updated model.

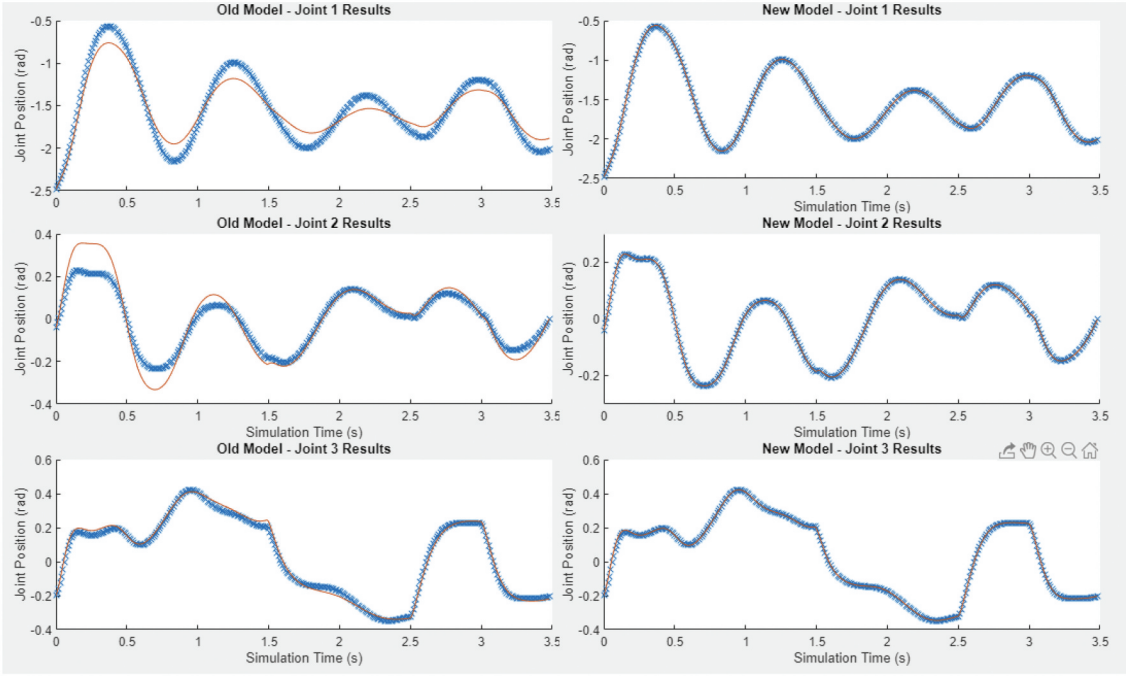


Figure 11. One of the resulting curve fits during the model variance testing, in which the Coulombic friction dynamics were present. The resulting curve fits were all greater than 97%.

shows the resulting curve fit in the 'New Model' side of the module's GUI. While the resulting model has parameter errors, and has determined non-existent dynamics to be present, the curve fits are excellent. Each resulting fit is greater than 97% by a normalised root mean square error metric. Thus, even though the model is incorrect, it is still capable of reflecting the true system.

These results establish that the use of grey-box system identification produces a high-fidelity model which is capable of representing the system. However, that model is incapable of perfectly determining the parameters of interest. This latter aspect may not be an issue for a DT application, but that is dependent on the application. Building on the results of this section, we now look to the other modules of the DT software to evaluate how effectively models produced by the approach and emulate a system.

6. System simulation module

6.1. Testing methodology

The System Simulation Module is responsible for applying the model determined by the System Identification Module for control design purposes. This module is designed to let a user simulate the motion of the manipulator using different controller gains, varying target times, and apply different DT models.

This particular module allows us to further evaluate the efficacy of the grey-box modelling process by comparing emulated motions from the DT with measured motions of the manipulator. The accuracy of the emulated motion can be calculated using a normalised root mean square error metric, providing a numerical value associated with the accuracy.

Evaluating this model, however, requires consideration of the closed loop dynamics of the system. Here, that leads to considerations about the computed torque controller. These controllers have integral terms that allow them to mitigate model errors, as many feedback controllers have, and this has an effect on the final trajectory. As such, the controller has the capacity to eliminate errors associated with modelling.

Additionally, it must be recognised that a computed torque controller is a type of model driven control. That is to say that it uses a believed model of the system to calculate the control signal in addition to the feedback loop.

Thus, two elements must be established to properly define this test. Recognizing the effect of the controller, different control target times must be used to evaluate the efficacy of the approach. Longer control target times will make it easier for the controller to force the manipulator to the desired path, increasing the accuracy of the simulation. Thus, by decreasing the control target time, we give more influence of the closed loop dynamics to the manipulator and thus can obtain insight into the efficacy of the DT model. In this work, we will test the module under control target times of 1 s, 2 s, 4 s, and 8 s. As these tests may produce control signals larger than ±8 V, the saturation limits are removed from both the module and simulated robotic manipulator.

Next, as a model is used by the controller in the calculation of the control signal, a model must be selected for use by the controller operating the manipulator and for the module itself. This opens another dimension of evaluating the efficacy of the approach as it will not only consider the effects on the simulation outcome from the module, but the effect of the model selection on the controller. For these tests, we choose to use a result from noise testing with 0.01 rad, which is presented in Figure 12. All curve fits in this test are greater than 91%, which represents a model that would more likely be encountered in a real world scenario compared to a model obtained with no noise, but it still under performing compared to most models that would be selected for a digital twin. Thus, it becomes a suitable choice for the testing conducted here, and one that tests the limits of the modelling approach.

6.2. Results and discussion

The resulting curve fits from all control target times, for each individual link, are presented in Figure 13. As anticipated, a lower control target time and accompanying more aggressive manoeuvre leads to a worsening curve fit. Conversely, higher control target times produce excellent results.

When more realistic control target times are used, the module demonstrates itself as an effective tool for virtualization purposes. Curve fits push past 97% in control target times greater than four seconds in all links, and such a curve fit is useful for the purposes of control design. This result supports the use of grey-box modelling for digital twins.

The results at lower control target times require more examination though. To assist in this discussion, a comparison of the emulated and manipulator

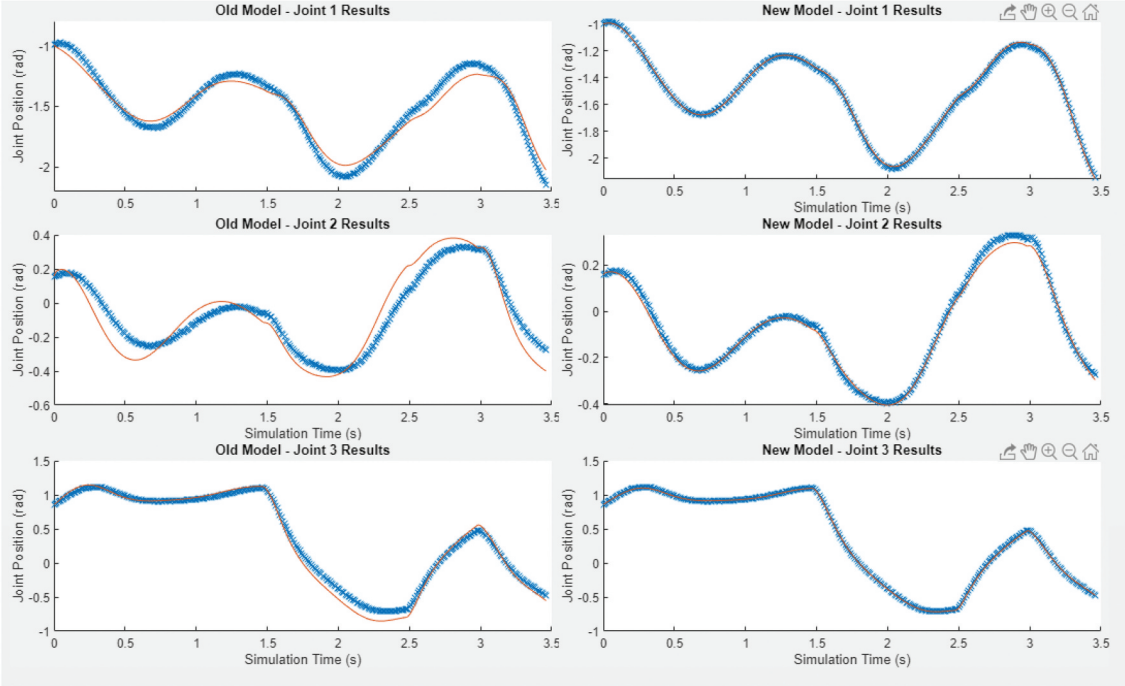


Figure 12. One of the resulting curve fits obtained during noise testing with 0.01 rad magnitude. All curve fits are greater than 91%.

response for the first link with a one second control target time is provided in Figure 14. Of note in these responses is that the actual manipulator response is smoother than the emulated response from the module. One would expect the emulated response to be smoother, as the model used in the controller and the module's emulation are identical. This particular behaviour may be explained by the different damping constants in the grey-box model, as that manipulator model very sensitive to changes in this parameter. As a result though, there are module's emulation becomes less accurate.

These results demonstrate some of the limitations in using grey-box system identification. When operating under extreme conditions, the errors in the determined dynamics of the manipulator push the emulation into a region where it no longer provides and accurate virtual representation of the system. However, under more realistic operating conditions the grey-box modelling approach produces an accurate representation which is suitable for this application. Whether the approach is deemed acceptable for use, however, is dependent on the application at hand.

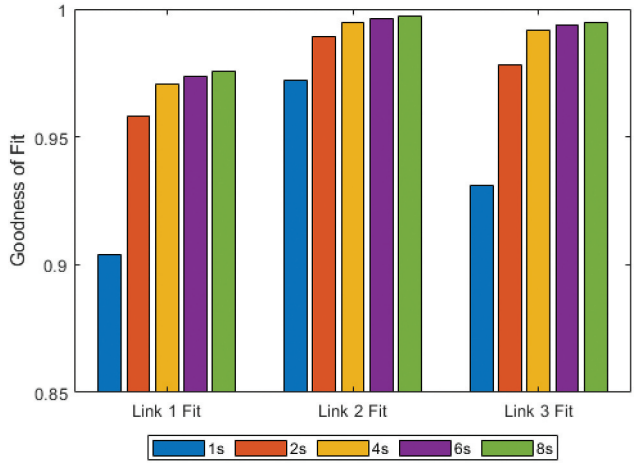


Figure 13. The resulting curve fits obtained during testing the System Simulation Module.

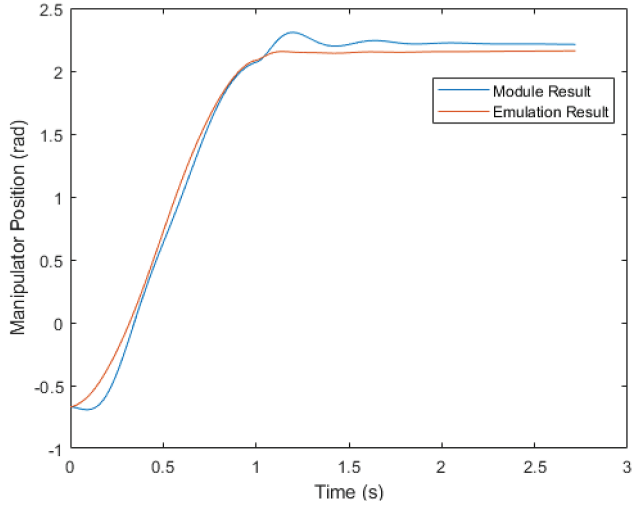


Figure 14. The resulting motion for the first link from the module and from the manipulator under a one second control target time.

7. System monitoring module

7.1. Testing methodology

The System Monitoring Module applies the grey-box model to determine if the system has experienced an anomaly. It uses three metrics to determine this, which are an absolute error, relative error, and number of consecutive errors. Only after all three of these have passed out of their tolerable limits does the module alert that a failure has occurred.

The absolute error is provided in Eq. 8 while the relative error is provided in Eq. 7. Here $\widehat{\theta}_i$ is the calculated position of the link from the module for link i, while θ_i is the measured position from the rotary encoder. These two are used in combination because at very small measured positions from the rotary encoder, such as those near 0 rad, the relative error will rapidly grow. To mitigate this from setting off a warning message, the absolute error is also considered. When both errors are outside their tolerable limits, the number of consecutive errors increases by one and a warning message is provided. When the number of consecutive errors exceeds its limit, then the failure message is given. However, if the errors return to within their tolerable limits before the consecutive error counter exceeds its limit, then the counter is reset. This feature prevents measurement errors from sources like sensor noise from falsely alerting the operator of an error.

$$e_{rel} = \frac{\left|\theta_i - \widehat{\theta}_i\right|}{\widehat{\theta}_i} \tag{7}$$

$$e_{abs} = \left|\theta_i - \widehat{\theta}_i\right| \tag{8}$$

Testing the use of a grey-box model in such an application provides insight into the efficacy of the model in a real-life application. Should the module successfully detect a failure, it will demonstrate that the grey-box modelling approach can find use in production-line applications.

Fully outlining this test requires several aspects to be defined. For the module itself, these include the model used in the test, the absolute and relative error bounds, the number of consecutive deviations required to alert of a failure, and the sampling frequency. On the side of the manipulator itself, the simulation parameters and type of failure must be defined.

In this test, we use a model from the 0.001 rad magnitude noise testing. The System Identification Module GUI result is shown in Figure 15. All curve fits here are greater than 98%, which is a significantly better fit compared to the model used in the System Simulation Module testing. This is done here to minimise deviations that would occur due to modelling errors, and rather test the approach under conditions that would be more likely acceptable for use in a digital twin.

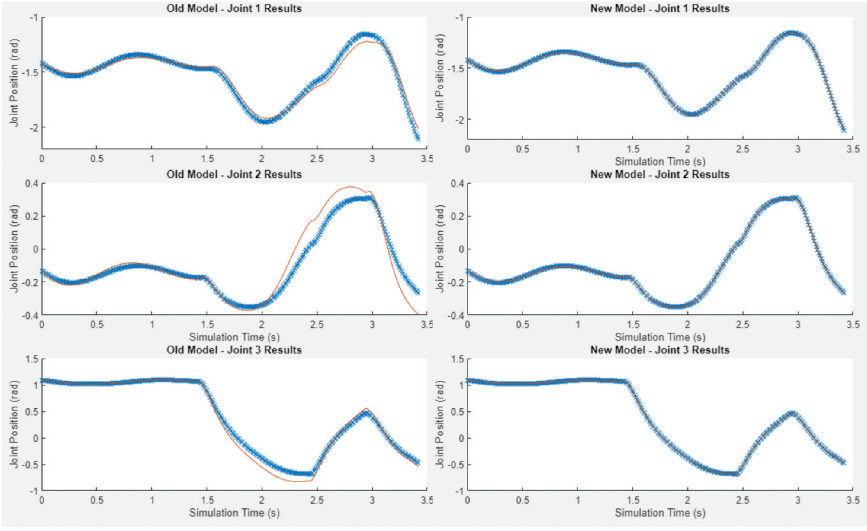


Figure 15. The model from the noise testing which is used in testing with the system monitoring module.

Additionally, the following parameters are given for the module. The limit of the absolute error is set to 0.10 rad, the limit of the relative error is set to 0.05 rad, and the maximum number of deviations is set to 10 before the module alerts the user of a failure. Lastly, the sampling period is set to 0.5s.

For the manipulator simulation, we force a failure in the second link in the form of an inability for the servo motor to convert voltage into torque as expected. At a simulation time of 45s, we reduce the voltage constant in the second servo to 10% of its true value. With the previously mentioned saturation limit of ±8 V, the controller will be unable to compensate for the growing error and the module should detect the deviation.

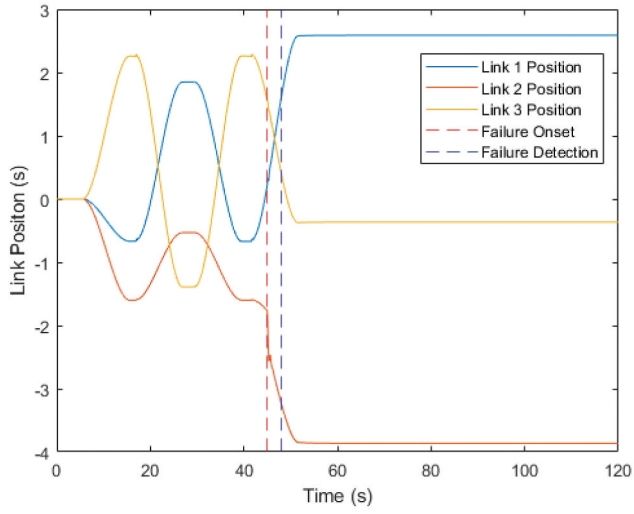


Figure 16. The results from testing the failure detection test, with the failure onset and detection labelled.

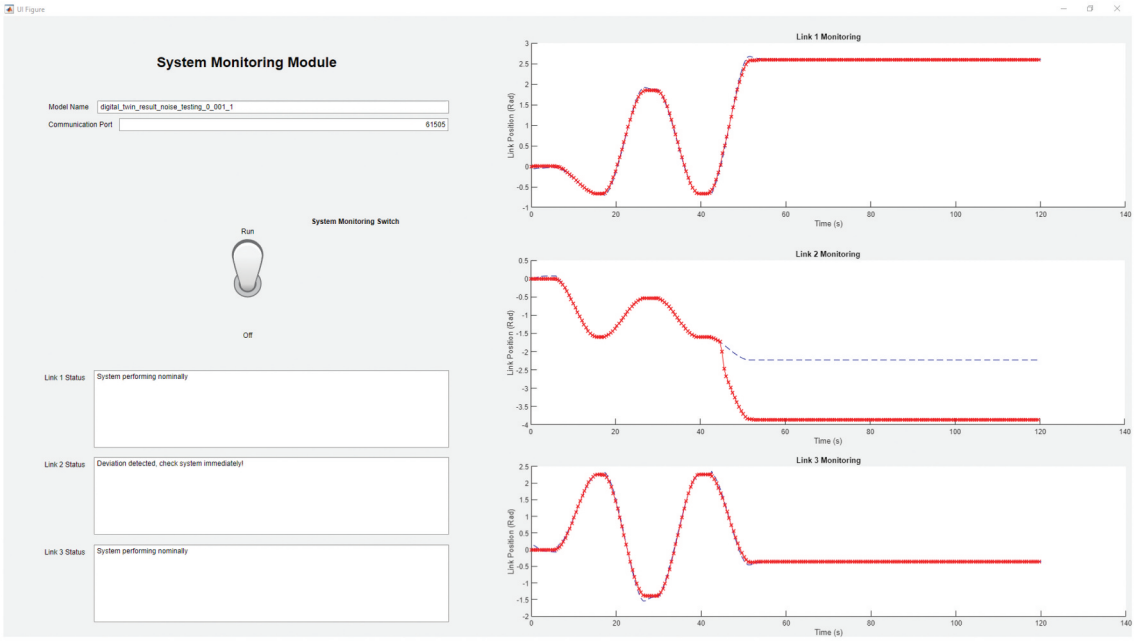


Figure 17. The module's GUI during the test. Note that the second window is displaying a message stating that failure in the second link has been detected.

7.2. Results and discussion

We present two results for the test in Figures 16 and 17. The module demonstrates itself as capable of detecting the failure in the second servo motor using this proposed technique. A consequence of the proposed methodology, however, is that it is incapable of detecting the mode of failure. As such, the module only informs the user that a failure occurred.

The results demonstrate an applicable use case for the DT model, and demonstrates that the grey-box system identification approach can produce a model accurate enough to provide system monitoring functions. Under the conditions and failure mode here, which was catastrophic, there is little issue with detecting that failure has occurred. The result here supports the use of grey-box system identification as a method for constructing models for digital twins.

The accuracy of the model does produce limitations though. As seen in the System Simulation Module, the controller was able to force the manipulator to the desired trajectory well enough that errors in the overall dynamics became minimal. However, that means there are smaller deviations which may be indicative of a fault developing could be missed. One such way of this occurring is the use of consecutive deviations being an indicator for failure. If this number is set very high, deviations that occurred which the controller can compensate for will be missed. Similarly, it also leads to larger error tolerances, as smaller tolerances would produce a warning when one is not necessary.

The solution to the previous issue is to employ more rigorous data metrics when determining errors. This, however, comes with the challenge of greater computational power requirements which may limit the ability to collect data consistently.

Despite the challenges and limitations that the model accuracy produces, this module still supports the use of grey-box models in system monitoring applications. With further refinement, this module may be able to provide more detailed observations on the system using the grey-box model.

8. Discussion

Despite the extensive literature on the application of digital twins to industrial applications (Abisset-Chavanne et al. 2024) there has been limited work on evaluating the performance of alternative modelling techniques to support system identification, system simulation, and system monitoring. Table 2 provides a list of recent studies on the application of data-driven and grey-box modelling techniques that include performance assessments of the digital twin model. As can be seen from this table, there isn't an agreed-upon standard test case for this work, and a result, the application areas vary considerably.

From an application perspective, the closest studies to the work presented in this paper are the papers by Borangiu et al. (2020) and Q. Yu-Ming, X. Bing, and D. San-Peng (2020) who apply digital twin technology to a pick-and-place robot. Only Borangiu et al. (2020) report on the model's performance; however in this case, the application relates to robot pick-and-place operation classification rather than position control (as reported in this paper). Their work does, however, illustrate the limitations of data-driven models noted previously: i.e. high model accuracy for operations within the training set, but low accuracy for abnormal operations.

This is further illustrated by the three additional data-driven modelling studies listed in Table 2. He et al. (2019) use a combined physics-based and data-driven model for chemical process measurement and estimation. They note 'For complex fault scenarios, the process diagnostic performance of digital twins is slightly insufficient, so the specific fault location needs to be further judged by online

Table 2. Digital twin modelling performance evaluations.

Author(s)	Model	Application
Borangiu et al. (2020)	Data-driven	robot pick-and-place
He et al. (2019)	Data-driven	chemical process measurement & estimation
D. Preuveneers, W. Joosen, and E. Ilie-Zudor (2018)	Data-driven	print and label manufacturing
Qamsane et al. (2021)	Data-driven	bearing fault detection
E. Atasayan, E. Milanov, and A. Dursun Alkan (2024)	Grey-box	ship manoeuvring dynamics
Liu et al. (2023)	Grey-box	optical network control & management
Morkunaite et al. (2024)	Grey-box	building thermal dynamics

instruments'. D. Preuveneers, W. Joosen, and E. Ilie-Zudor (2018) use a data-driven model to detect errors in a 'print and label' manufacturing process. The digital twin showed very good performance in most cases (it was wrong in 10% of the cases reported), but the authors note that further training is required to generalise the results. Finally, Qamsane et al. (2021) use a data-driven model for bearing fault detection. Their case study focuses on a simplified model (only vibration data in a univariate model) that has limitations for the assessment of a condition monitoring digital twin.

Presently, this study is unique in that it reports on the application and performance assessment of grey-box system identification techniques for digital twins in manufacturing automation. Although considerable work has been conducted on the application of digital twins to manufacturing automation (as reported in Section 2), the majority of the studies have focused on establishing a general and unifying definition of the industrial digital twin, establishing a consensual architecture for the digital twin, and characterising the potential environmental impact of digital twin use cases (Abisset-Chavanne et al. 2024). However, some comparisons can be made with similar studies outside the manufacturing automation domain.

E. Atasayan, E. Milanov, and A. Dursun Alkan (2024) propose a grey-box model to investigate the feasibility of determining ship manoeuvring dynamics using system identification methods, and to evaluate the manoeuvring characteristics using experimental results compared to predicted results. They note that the model effectively predicts the manoeuvring performance. Although we cannot make direct comparisons between our test cases, E. Atasayan, E. Milanov, and A. Dursun Alkan (2024) show similar results to those reported in this paper in terms of parameter estimation and model accuracy.

Liu et al. (2023) propose a grey-box model of optical-network capacity to support more efficient network control and management. Unlike the data-driven modelling studies listed in Table 2, their model can be generalised to unseen scenarios since it is based on the underlying physics of the physical system. This is consistent with the observations reported in Section 6 of our proposed grey-box model's ability to operate over a wide range of realistic operating conditions.

Finally, Morkunaite et al. (2024) propose a grey-box modelling approach for multi-story residential building thermal dynamics. Their rationale for the use of a grey-box modelling approach is consistent with that reported in this paper: commonly used physics-based white box models are time-consuming and require vast expertise, while black box forecasting models, which rely primarily on building energy consumption data, lack fundamental insights and hinder re-use. Although direct comparisons cannot be made between our test cases, they report similar results in terms of parameter estimation and model accuracy. More specifically, despite some inconsistencies in parameter estimates, a high-fidelity model is produced which is capable of representing the underlying physical system.

9. Conclusions & future work

In this paper we evaluated the use of grey-box system identification as a means for constructing a high-fidelity model for use in DTs. We defined fidelity as the level of detail a model contains about the underlying system. Grey-box modelling provides a route to achieve a high-fidelity model without the need for massive amounts of data, but still accounts for data collected through sensors. This enables the approach to be more accurate than white-box modelling, while not being as computationally expensive as a black-box model.

The results of using grey-box system identification demonstrate that the approach is incapable of perfectly estimating the parameters of the system. We performed testing where external disturbances, sensor noise, and model uncertainty were accounted for. In all cases, the approach was unable to perfectly determine the parameters, even in cases where the sensor data was perfect.

When viewing the results in the form of GTV, however, the approach demonstrates itself as a useful method for creating a model which effectively explains the behaviour of the system. The produced models, even in the presence of sensor noise, modelling error, and external disturbances, produced excellent curve fits which accurately simulate the system. This result suggests that the approach, with its limitations, may still be effective for use in DTs.

The tests with the System Simulation Module and Monitoring Module support the efficacy of the models created using grey-box system identification for use in DTs. The models, when used in combination with a

feedback controller, demonstrate accurate simulation capabilities under normal operating conditions. Similarly, these models are also useful in system monitoring tasks like failure detection. However, as the demands of the system become more intense the modelling errors play a larger role and reduce the overall efficacy of the determined model. Thus, there are limits as to where the model is useful.

Our future work aims to further evaluate the models produced in monitoring applications. Currently, a very rudimentary approach is used to determine catastrophic failures. But this leads to gaps in the System Monitoring Module's capabilities in providing predictive maintenance as well as determining failure modes. Addressing this will further test the limits of the grey-box system identification modelling approach, and provide more insight into the capabilities of our DT architecture.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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References

Abisset-Chavanne, E., T. Coupaye, F. R. Golra, D. Lamy, A. Piel, O. Scart, and P. Vicat-Blanc. 2024. "A Digital Twin Use Cases Classification and Definition Framework Based on Industrial Feedback." *Computers in Industry* 161:104113. https://doi.org/10.1016/j.compind.2024.104113.

Ait-Alla, A., M. Kreutz, D. Rippel, M. Lütjen, and M. Freitag. 2019. "Simulation-Based Analysis of the Interaction of a Physical and a Digital Twin in a Cyber-Physical Production System." *IFAC Papersonline* 52 (13): 1331–1336. https://doi.org/10.1016/j.ifacol.2019.11.383.

Atasayan, E., E. Milanov, and A. Dursun Alkan. 2024. "Application of an Offline Grey Box Method for Predicting the Manoeuvring Performance." *Brodogradnja* 75 (3): 1–36. https://doi.org/10.21278/brod75304.

Azangoo, M., A. Taherkordi, and J. O. Blech. (2020). "Digital Twins for Manufacturing Using UML and Behavioral Specifications." *IEEE International Conference on Emerging Technologies and Factory Automation*, Vienna, Austria, 1035–1038.

Biesinger, F., D. Meike, B. Kraß, and M. Weyrich. 2019. "A Digital Twin for Production Planning Based on Cyber-Physical Systems: A Case Study for A Cyber-Physical System-Based Creation of A Digital Twin." *Procedia CIRP* 79:355–360. https://doi.org/10.1016/j.procir.2019.02.087.

Bohlin, T., and S. Graebe (1994). "Issues in Nonlinear Stochastic Grey-Box Identification." *IFAC Proceedings* 27 (2): 1175–1180.

Borangiu, T., S. Raileanu, A. Silisteanu, S. Anton, and F. Anton (2020). "Smart Manufacturing Control with Cloud-Embedded Digital Twins." *24th International Conference on System Theory, Control and Computing*, Sinaia, Romania, 915–920.

Grieves, M., and J. Vickers. 2017. "Digital Twin: Mitigating Unpredictable, Undesirable Emergent Behavior in Complex Systems." In *Transdisciplinary Perspectives on Complex Systems*, edited by Franz-Josef Kahlen, Shannon Flumerfelt, and Anabela Alves, 85–113. Switzerland: Springer International Publishing.

He, R., G. Chen, C. Dong, S. Sun, and X. Shen. 2019. "Data-Driven Digital Twin Technology for Optimized Control in Process Systems." *ISA Transactions* 95:221–234. https://doi.org/10.1016/j.isatra.2019.05.011.

IBM (2022). *What Is a Digital Twin?*" Accessed February 28, 2023. https://www.ibm.com/topics/what-is-a-digital-twin.

ISO (2021). "*Automation Systems and Integration — Digital Twin Framework for Manufacturing Part 1: Overview and General Principles*." International Organization for Standardization Accessed: 2024-07-10. https://www.iso.org/standard/75066.html.

Jeon, S. M., and S. Schuesslbauer. (2020). "Digital Twin Application for Production Optimization." In *IEEE International Conference on Industrial Engineering and Engineering Management*: 542–545. Singapore: IEEE Computer Society.

Kristensen, N. R., H. Madsen, and S. B. Jørgensen. 2004. "Parameter Estimation in Stochastic Grey-Box Models." *Automatica* 40 (2): 225–237. https://doi.org/10.1016/j.automatica.2003.10.001.

Kritzinger, W., M. Karner, G. Traar, J. Henjes, and W. Sihn. 2018. "Digital Twin in Manufacturing: A Categorical Literature Review and Classification." *IFAC Papersonline* 51 (11): 1016–1022. https://doi.org/10.1016/j.ifacol.2018.08.474.

Leng, J., H. Zhang, D. Yan, Q. Liu, X. Chen, and D. Zhang. 2019. "Digital Twin-Driven Manufacturing Cyber-Physical System for Parallel Controlling of Smart Workshop." *Journal of Ambient Intelligence and Humanized Computing* 10 (3): 1155–1166. https://doi.org/10.1007/s12652-018-0881-5.

Lesage, J., and R. Brennan. (2022). "Constructing Digital Twins for IEC61499 Based Distributed Control Systems." *IEEE International Conference on Systems, Man, and Cybernetics* (*SMC*), Prague, Czech Republic, 1834–1839.

Liu, Y., X. Liu, Y. Zhang, M. Cai, M. Fu, X. Zhong, L. Yi, W. Hu, and Q. Zhuge. 2023. "Building a Digital Twin of an EDFA for Optical Networks: A Gray-Box Modeling Approach." *Journal of Optical Communications and Networking* 15 (11): 11):830–838. https://doi.org/10.1364/JOCN.499530.

Liu, Z., W. Chen, C. Zhang, C. Yang, and Q. Cheng. 2020. "Intelligent Scheduling of a Feature-Process-Machine Tool Supernetwork Based on Digital Twin Workshop." *Journal of*

Manufacturing Systems 58:157–167. https://doi.org/10.1016/j.jmsy.2020.07.016.

Meier, N., R. Müller-Polyzou, L. Brach, and A. Georgiadis. 2021. "Digital Twin Support for Laser-Based Assembly Assistance." *Procedia CIRP* 99:460–465. https://doi.org/10.1016/j.procir.2021.03.066.

Morkunaite, L., J. Kardoka, D. Pupeikis, P. Fokaides, and V. Angelakis. 2024. "Digital Twin for Grey Box Modeling of Multistory Residential Building Thermal Dynamics." *Arxiv* 2402 (2909): 1–6.

Preuveneers, D., W. Joosen, and E. Ilie-Zudor. (2018). "Robust Digital Twin Compositions for Industry 4.0 Smart Manufacturing Systems." *IEEE International Enterprise Distributed Object Computing Workshop*, Stockholm, Sweden, 69–78.

Qamsane, Y., J. Moyne, M. Toothman, I. Kovalenko, E. C. Balta, J. Faris, D. M. Tilbury, and K. Barton. 2021. "A Methodology to Develop and Implement Digital Twin Solutions for Manufacturing Systems." *IEEE Access* 9:44247–44265. https://doi.org/10.1109/ACCESS.2021.3065971.

Reynders, G., J. Diriken, and D. Saelens. 2014. "Quality of Grey-Box Models and Identified Parameters as Function of the Accuracy of Input and Observation Signals." *Energy and Buildings* 82:263–274. https://doi.org/10.1016/j.enbuild.2014.07.025.

Schleich, B., N. Anwer, L. Mathieu, and S. Wartzack. 2017. "Shaping the Digital Twin for Design and Production Engineering." *CIRP Annals - Manufacturing Technology* 66 (1): 141–144. https://doi.org/10.1016/j.cirp.2017.04.040.

Talkhestani, B. A., D. Braun, W. Schloegl, and M. Weyrich. 2020. "Qualitative and Quantitative Evaluation of Reconfiguring an Automation System Using Digital Twin." *Procedia CIRP* 93:268–273.

Tao, F., Q. Qi, L. Wang, and A. Y. Nee. 2019. "Digital Twins and Cyber–Physical Systems toward Smart Manufacturing and Industry 4.0: Correlation and Comparison." *Engineering* 5 (4): 653–661. https://doi.org/10.1016/j.eng.2019.01.014.

Tao, F., B. Xiao, Q. Qi, J. Cheng, and P. Ji. 2022. "Digital Twin Modeling." *Journal of Manufacturing Systems* 64:372–389. https://doi.org/10.1016/j.jmsy.2022.06.015.

Thelen, A., X. Zhang, O. Fink, Y. Lu, S. Ghosh, B. D. Youn, M. D. Todd, S. Mahadevan, C. Hu, and Z. Hu. 2022. "A Comprehensive Review of Digital Twin — Part 1: Modeling and Twinning Enabling Technologies." *Structural and Multidisciplinary Optimization* 65 (12): 354.

Tulleken, H. J. 1993. "Grey-Box Modelling and Identification Using Physical Knowledge and Bayesian Techniques." *Automatica* 29 (2): 285–308. https://doi.org/10.1016/0005-1098(93)90124-C.

Uhlemann, T. H., C. Lehmann, and R. Steinhilper. 2017. "The Digital Twin: Realizing the Cyber-Physical Production System for Industry 4.0." *Procedia CIRP* 61:335–340. https://doi.org/10.1016/j.procir.2016.11.152.

Vachalek, J., L. Bartalsky, O. Rovny, D. Sismisova, M. Morhac, and M. Loksik. (2017). "The Digital Twin of an Industrial Production Line within the Industry 4.0 Concept." *Proceedings of the 2017 21st International Conference on Process Control*, Štrbské Pleso, Slovakia, 258–262.

Wu, C., Y. Zhou, M. V. P. Pessˆoa, Q. Peng, and R. Tan. 2021. "Conceptual Digital Twin Modeling Based on an Integrated Five-Dimensional Framework and Triz Function Model." *Journal of Manufacturing Systems* 58:79–93. https://doi.org/10.1016/j.jmsy.2020.07.006.

Xia, L., J. Lu, and H. Zhang. (2020). "Research on Construction Method of Digital Twin Workshop Based on Digital Twin Engine." *Proceedings of 2020 IEEE International Conference on Advances in Electrical Engineering and Computer Applications*, Dalian, China, 417–421.

Yu-Ming, Q., X. Bing, and D. San-Peng. (2020). "Research on Intelligent Manufacturing Flexible Production Line System Based on Digital Twin." *35th Youth Academic Annual Conference of Chinese Association of Automation*, Zhanjiang, China, 854–862.

Zhang, K., T. Qu, D. Zhou, H. Jiang, Y. Lin, P. Li, H. Guo, Y. Liu, C. Li, and G. Q. Huang. 2020. "Digital Twin-Based Opti-State Control Method for a Synchronized Production Operation System." *Robotics and Computer-Integrated Manufacturing* 63 (101892): 101892. https://doi.org/10.1016/j.rcim.2019.101892.

Zhao, R., D. Yan, Q. Liu, J. Leng, J. Wan, X. Chen, and X. Zhang. 2019. "Digital Twin-Driven Cyber-Physical System for Autonomously Controlling of Micro Punching System." *IEEE Access* 7:9459–9469. https://doi.org/10.1109/ACCESS.2019.2891060.

Zhu, J., J. Zhang, Q. Zhu, L. Zuo, C. Liang, X. Chen, and Y. Xie. 2024. "Virtual Geographical Scene Twin Modeling: A Combined Data-Driven and Knowledge-Driven Method with Bridge Construction as a Case Study." *International Journal of Digital Earth* 17 (1): 1):1–23. https://doi.org/10.1080/17538947.2024.2356126.

Zhu, Z., C. Liu, and X. Xu. 2019. "Visualisation of the Digital Twin Data in Manufacturing by Using Augmented Reality." *Procedia CIRP* 81:898–903. https://doi.org/10.1016/j.procir.2019.03.223.

Zhuang, C., T. Miao, J. Liu, and H. Xiong. 2021. "The Connotation of Digital Twin, and the Construction and Application Method of Shop-Floor Digital Twin." *Robotics and Computer- Integrated Manufacturing* 68 (4): 102075. https://doi.org/10.1016/j.rcim.2020.102075.

Appendix

Appendix A. Physical properties of the robotic manipulator

The underlying physics of the robotic manipulator shown in Figure 1 are described in Section 3 and (Lesage and Brennan 2022). Table A1 summarises the relevant physical properties of the manipulator.

Table A1. Physical properties of the robotic manipulator

Property	Symbol	Value
Link 1 Moment of Inertia	I_{z1}	5.3325×10^{-4} kgm^2
Link 2 Moment of Inertia	I_{z2}	3.7766×10^{-4} kgm^2
Link 3 Moment of Inertia	I_{z3}	2.4296×10^{-4} kgm^2
Gripper Inertia	$I_{gripper}$	0.1 kgm^2
Link 1 Mass	m_1	0.11804 kg
Link 2 Mass	m_2	0.11604 kg
Link 3 Mass	m_3	0.12994 kg
Link 1 Length	l_1	0.1578 m
Link 2 Length	l_2	0.1325 m
Link 3 Length	l_3	0.1608 m
Link 1 Centre of Mass	cg_1	0.120 m
Link 2 Centre of Mass	cg_2	0.100 m
Link 3 Centre of Mass	cg_3	0.140 m
Voltage Constant 1	K_{t1}	0.2533 N m/V
Voltage Constant 2	K_{t2}	0.1573 N m/V
Voltage Constant 3	K_{t3}	0.1573 N m/V
Damping Constant 1	D_1	0.05 N m/rad/s
Damping Constant 2	D_2	0.05 N m/rad/s
Damping Constant 3	D_3	0.05 N m/rad/s